ZYK-563 · OCTOBER 24, 2026 · THE FULL SOLVE

THE HITCHHIKER WATCH - RECLAMATION PHASE 2

byline CAT ROBINSON-RUTELLA (business surface) · her hand fires the submission · this paper is the solve, the submission walkthrough lives on the roadmap site

> *Phase 1 is closed and that is fine: the Bureau of Reclamation's "Halt the Hitchhiker" Phase 2 window has not opened yet. This paper is the solve built and waiting, so the submission fires the week the window opens, not the month after.*

The door

Fact	Detail
Prizes	$125,000 / $75,000 / $50,000 in the final phase structure
Problem	Invasive mussels (quagga/zebra) transported in boat ballast water; decontamination that actually gets used at boat ramps
Phase 1 status	Concept window CLOSED; Phase 2 to be announced - the watch, monitored via haltthehitchhiker.yet2.com and usa.gov/find-active-challenge

The solve: deterministic decontamination verification

Every current approach pairs a treatment (hot water, filtration, drying time) with a promise. The pitch is the same shape as the clinical documentation pitch: the treatment is only real if the verification is deterministic.

- **Sensor-fusion counting:** pre- and post-treatment veliger density read from the water column with paired optical + conductivity sensors, counted deterministically - no model votes on whether a mussel is there, a threshold does, and the threshold is published.
- **Dose calibration as arithmetic:** temperature and exposure time are computed from measured flow rate and volume, shown as a printed equation on the unit's face. A ramp operator can check the math by hand.
- **The record:** every decontamination event presses to text - boat ID hash, in/out counts, dose delivered, pass/fail - the LedgerLine audit-trail architecture in a new body.
- **The cheap part wins:** the judges fund adoption. Our unit prices at trailer-ramp economics because the intelligence is a rule file, not a GPU.

Watch procedure: Monday checks on the Phase 2 announcement (ride with the rip reminder, Mon 11 AM). When the window opens: registration, the solve paper becomes the entry, her hand fires.

ThreadCat public baseline (threadcat.org) holds the licensed method. Sources pulled live Oct 24, 2026. Nothing sent outward: her hand only.